

A Disproof of Conjecture 00000008549

gaochengzhecpu (AI-assisted submission)

The conjecture's assertion that the diamond lattice is a smallest partition lattice with variable maximal-chain length is false. The diamond M_3 is the partition lattice Π_3 , and all of its maximal chains have length 2. Thus it is not an example with variable maximal-chain length at all, irrespective of minimality.

Precise scope. A partition is a collection of nonempty disjoint blocks whose union is the underlying set. Partitions are ordered by refinement: $\pi \leq \sigma$ if every block of π is contained in a block of σ . Equivalently, the same-block equivalence relation of π is contained in that of σ . We use the standard five-element diamond M_3 , with bottom, top, and three incomparable atoms. The ordinary refinement order and the Π_3 diagram are also given in the MIT lecture notes [1]. The length of a nonempty finite chain is its number of elements minus one. A maximal chain is maximal under inclusion among all chains.

The source's English first clause concerns graded lattices, whereas its Chinese first clause says ordered lattices. This proof does not use or challenge that first clause. It disproves the second clause, which explicitly describes the diamond as an example with variable length in both languages.

All partitions on three points. Use the set $\{0, 1, 2\}$. A partition has either three, two, or one block. In the first and last cases it is forced. In the two-block case, one block has two elements, and there are exactly three choices for that block. Thus all partitions are

label	partition	rank
0	$\{\{0\}, \{1\}, \{2\}\}$	0
a	$\{\{0, 1\}, \{2\}\}$	1
b	$\{\{0, 2\}, \{1\}\}$	1
c	$\{\{1, 2\}, \{0\}\}$	1
1	$\{\{0, 1, 2\}\}$	2

Each two-block partition lies above the discrete partition and below the one-block partition. Distinct two-block partitions do not refine one another. This proves $\Pi_3 \cong M_3$ as ordered sets, and hence as lattices: least upper and greatest lower bounds are determined by the order.

All maximal chains. Let C be any maximal chain of this lattice. Both the bottom and top must belong to C , since either could otherwise be added to it. It contains at most one of a, b, c , since those are incomparable. It must contain at least one of them, because $\{0, 1\}$ can be extended by inserting any atom. Consequently the complete list of maximal chains is

$$0 < a < 1, \quad 0 < b < 1, \quad 0 < c < 1.$$

Each has exactly three elements and length 2. Conversely, each listed chain is maximal because its only missing elements are incomparable with its atom. The claimed variable-length example therefore does not exist in the diamond.

Formal verification and exhaustive scope. The self-contained Lean 4.19.0 file imports only `Std`. It encodes a binary relation on three points by its nine truth values in `Fin 512`, and imposes reflexivity, symmetry, and transitivity. The theorem `every_prop_partition_encoded` shows that every proposition-valued equivalence relation has such an encoding. Thus the formal definition of a partition as a same-block equivalence relation includes every actual partition.

The only equivalence-relation codes are 273, 283, 341, 433, 511, corresponding respectively to the five partitions displayed above. The maps `partitionOf` and `label` are proved inverse. The theorem `arbitrary_partition_refinement` proves that actual relation inclusion is precisely the diamond order. The lattice laws for this order and its meet and join are also verified.

All families of partitions, including arbitrary proposition-valued families, are uniquely encoded by five membership bits. `MaximalPartitionChain` quantifies over all such families when expressing maximality under inclusion. The proof transfers this definition to the complete set of 32 subset codes, where the only maximal-chain codes are 19, 21, 25 and each has cardinality 3. It also transfers maximality back and proves that these three actual maximal chains exist, so the length conclusion is not vacuous.

The theorem `every_maximal_partition_chain_length_two` establishes length 2 for every maximal chain. The final theorem `conjecture_8549_diamond_claim_false` negates the existence of two actual maximal chains with unequal lengths. This directly negates a necessary part of the conjecture’s claim that the diamond is a variable-length example. There are no admitted proofs or additional axioms; the final theorem uses only `propext` and `Quot.sound`.

References

- [1] A. Postnikov, *Algebraic Combinatorics, Lecture 9*, MIT 18.212, Spring 2021, section on the partition lattice. https://math.mit.edu/~apost/courses/18.212_2021/lectures/18212_lecture09.pdf